COM2020 Team Project

Lecture 3: Prototyping

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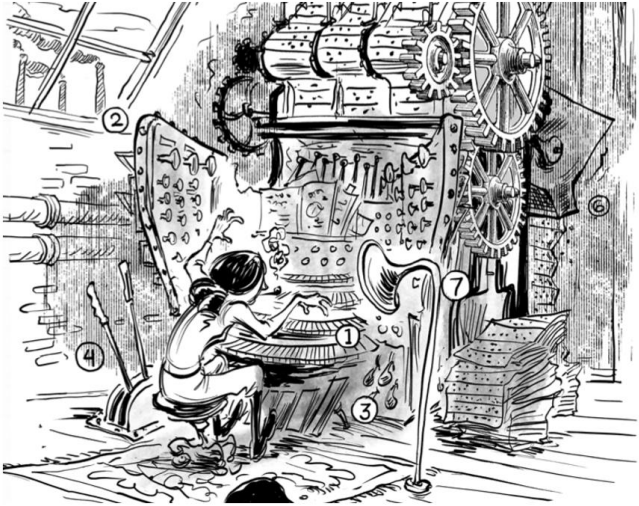
Module Vision

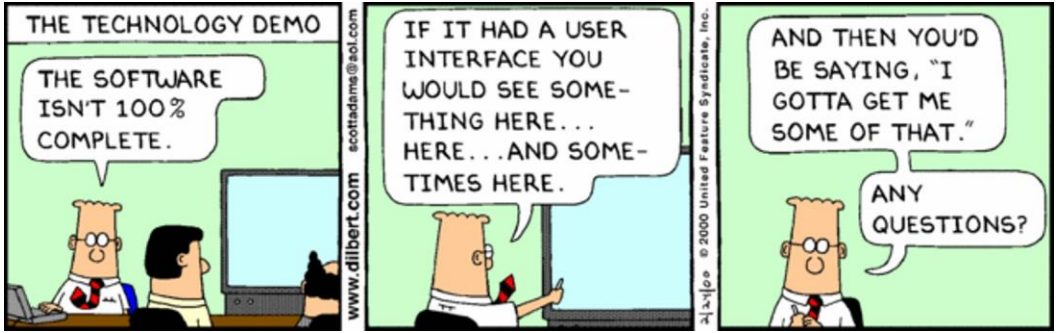


COM2020

Team Project

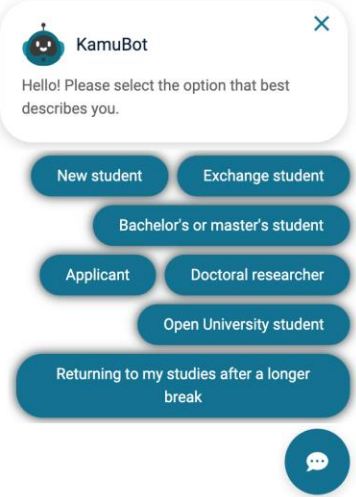
Teams apply technical skills in software development and data analysis while managing project planning, teamwork, and communication.

Prototyping



Prototyping - Problem

User reaction is difficult to predict



What is a prototype?

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A prototype is a concrete but partial representation or implementation of a system design.

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A prototype is one manifestation of a design that allows stakeholders to interact with it and to explore its suitability.

What is a prototype?

Prototypes are tools for learning (not mini-products)

- User Centric (User Centered)
- Experimental and partial design
- Testing design ideas
- Something to do early on

Why prototype?

A prototype is a purposefully incomplete artefact built to answer a question

Prototype to learn

- Validate assumptions
- Find usability issues
- Check data/model feasibility
- Explore edge cases

Prototype to communicate

- Discuss or evaluate ideas
- Get stakeholder buy-in
- Align on scope & flows
- Make requirements concrete
- Clarify what "done" means

Prototype to de-risk

- Measure performance
- Reduce rework later
- Inform your project plan

Prototyping

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- In our field of computer science, representations such as screen sketches and simple early prototypes blend into each other
- Prototype is interactive
- The appropriateness of a prototype will depend on a number of factors such as
    - whom the prototype is aimed at
    - the stage of the design process
    - what features the designer is looking to explore

Main Kinds of Prototyping

- There are two main kinds of prototyping
  - High-fidelity (Hi-fi)
  - Low-fidelity (Lo-fi)

The PalmPilot wooden prototype
Source: https://www.computerhistory.org. © Mark Richards

Hi-fi Prototypes

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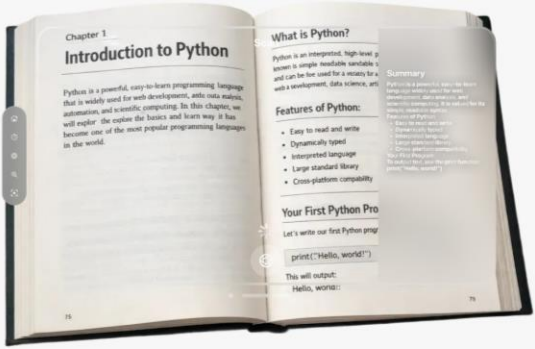
- Hi-fi prototypes are similar in look and feel to the anticipated final product (not necessarily in functionality).
- They are produced in software, whether in the development environment which will be used for implementation or in packages that will allow interactive effects to be mocked up easily.



A User-Centred Approach to Enhancing Engagement and Task Management for ADHD Students Using AR and Generative AI.
Ritul Parmar, Solomon Sunday Oyelere, Kehinde Aruleba

Hi-fi Prototypes

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Imikode: A VR Game to Introduce OOP Concepts

Figure 1: A. First game level. B. Environment updates throughout the game. C. Code updates throughout the game

Figure 2: A. Fox Instantiation. B. Fox walk behavior. C. Fox die behavior

Bouali, N., Nygren, E., Oyelere, S. S., Suhonen, J., & Cavalli-Sforza, V. (2019, November). Imikode: A VR game to introduce OOP concepts. In Proceedings of the 19th koli calling international conference on computing education research (pp. 1-2). https://doi.org/10.1145/3364510.3366149

Features of Hi-fi Prototypes

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- It is useful for detailed evaluation of the main design elements (content, visuals, interactivity, functionality and media) –
  - e.g. hi-fi prototypes can be used in usability studies to establish whether people can learn to use the system within a specified amount of time.
- It often constitutes a crucial stage in client acceptance.
- It is generally developed fairly well into the project when ideas are beginning to firm up, unless there is some crucial issue that needs to be resolved before any other work can proceed.

Some Considerations for Hi-fi's

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- A problem with developing hi-fi prototypes is that people believe them!
- Simple error – perhaps in the name of a customer, or of a product – can completely ruin a prototype
- Accurate detail is vital - It is no good saying ‘we were going to fix that’ or ‘that is just a place holder’
- A degree of effort and time is consumed in producing the prototype

Lo-fi Prototypes

Lo-fi prototypes, – often termed paper prototypes since that is what they are usually made from

Source: David Benyon

Features of Lo-fi Prototypes

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- They are more focused on the broad underlying design ideas – such as content, form and structure, the ‘tone’ of the design, key functionality requirements and navigational structure.

- They are designed to be produced quickly and thrown away as quickly.

- They capture very early design thinking and should aid, not hinder, the process of generating and evaluating many possible design solutions.

Considerations for Lo-fi's

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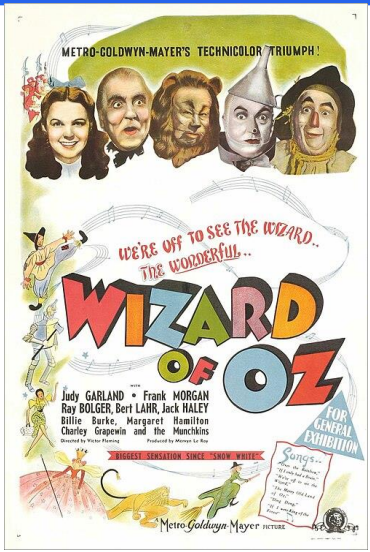
- The most usual form of lo-fi prototype is a series of ‘screenshots’ that people can ‘walk through’ (for example, a button on screenshot 1 can be ‘clicked’ and this is followed by screenshot 6, etc.).
- Wireframes implemented with a suitable software package can constitute a lo-fi prototype.
- How the prototype is implemented is limited only by your imagination, by time and the materials readily to hand.

Considerations for Lo-fi's

- Very flexible prototypes can be produced simply and quickly using screen-sized pieces of stiff paper and index cards or Post-its in different colours.
- Permanent features of each screen are drawn on the card; dynamic items such as dialogue boxes or menus use the cards or Post-its, cut to size as necessary.
- Do not to spend too much time doing this – the whole point is the low investment in the prototype build.

Lo-fi - Wizard of Oz Prototypes

- Powerful way to prototype as yet underdeveloped technologies.
- The method is so-called because in the 1953 film The Wizard of Oz, the seemingly powerful wizard with his booming voice is revealed to be a less terrifying character when Dorothy's dog pulls back a curtain.
- In UX design, the Wizard of Oz technique replaces some part of the technology with human intervention, so when a user interacts with the system, a person plays the role that the technology would do in the final operational system.



Source: https://en.wikipedia.org/wiki/The_Wizard_of_Oz

Wizard of Oz Prototypes

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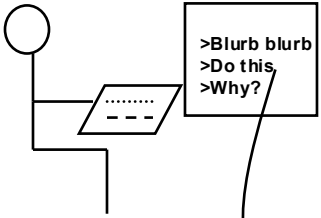
Facilitator

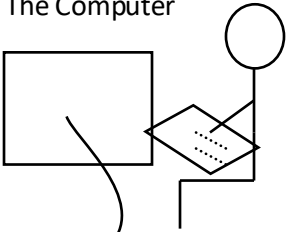
User



To test the prototype, you need three roles:

1. **The User:** Tries to complete a task (e.g., "Create an account").
2. **The Facilitator:** Explains the scenario but does not help the user.
3. **The Computer:** The person who "renders" the UI. If the user "taps" a paper button, the Computer physically places a new piece of paper (a menu or a new screen) in front of them.

The Computer



Lo-fi Prototyping - example

Wizard of Oz Prototyping

Advantages & Disadvantages of low-and high-fidelity prototypes

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Type	Advantages	Disadvantages
Low-fidelity prototype	•Quick revision possible •More time can be spent on improving the design before starting development •Evaluates multiple design concepts •Useful communication device •Proof of concept	•Limited error checking •Poor detailed specification for development •Facilitator-driven •Limited usefulness for usability tests •Navigational and flow limitations
High-fidelity prototype	•(Almost) complete functionality •Fully interactive •User-driven •Clearly defines navigational scheme •Use for exploration and test •Look and feel of intended product •Serves as a “living” or evolving specification •Marketing and sales tool	•More resource-intensive to develop •Time-consuming to modify •Inefficient for proof-of-concept designs •Potential of being mistaken for the final product •Potential of setting inappropriate expectations

Paper Prototyping

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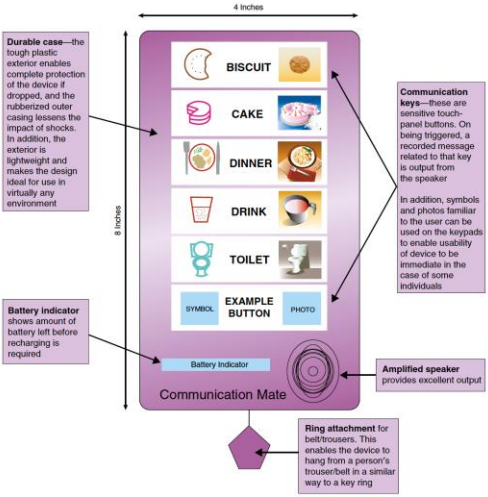
- A **disposable**, hand-drawn representation of a software interface.
- It is used to observe how a user interacts with a concept before any digital assets are created.
- Used by telephone and tablet companies as a core part of their design process

Feature-rich: megapixel cameras, music players, media galeries, downloaded applications, etc.

Prototype developed for cell phone user interface

Paper Prototyping Example



A paper-based prototype of a handheld device to support an autistic child. Source: Interaction design 5th Ed

Video Prototypes

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- Video prototyping is a powerful technique in computer science and software engineering where you create a short film to simulate how a system, interface, or user experience will function.
- It acts as a bridge between abstract requirements and technical implementation.

Benefits of Video Prototypes

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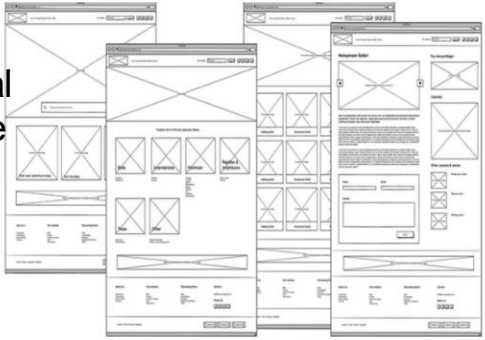
- Bridging the communication gap
  - Visualising the invisible
  - Alignment
- Cheap and fast
- Can serve as spec for developers

Wireframes

A wireframe is a lo-fi, two-dimensional skeletal outline of a webpage or app interface. Think of it as the "architectural blueprint" before the UI designers move in.

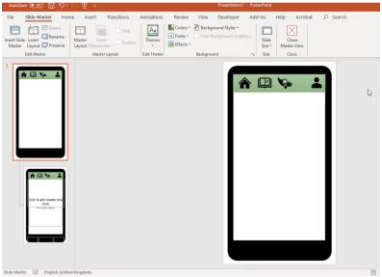
- Focus: Functionality, behavior, and hierarchy.
- Non-Focus: Colors, graphics, or complex typography.



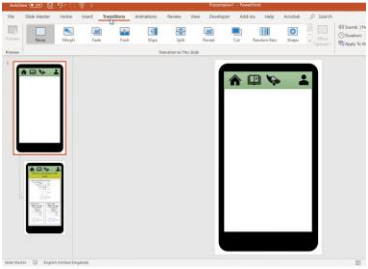
Slidedeck

It is a mid-to-high fidelity non-functional mockup that uses slide transitions and hyperlinks to mimic a user interface. It’s the "illusion" of software.

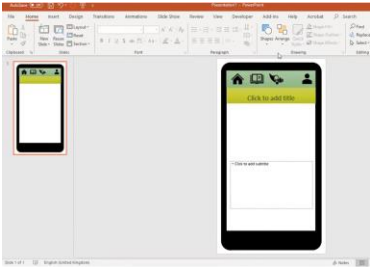
- The Mechanism: Every slide represents a specific state of the app.
- The Interaction: Buttons are actually "transparent shapes" with hyperlinks that jump the viewer to a different slide.



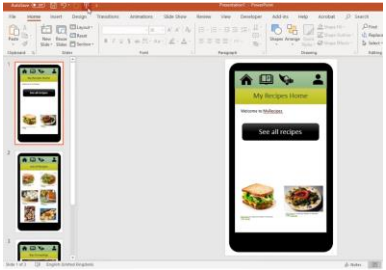
Create individual layout

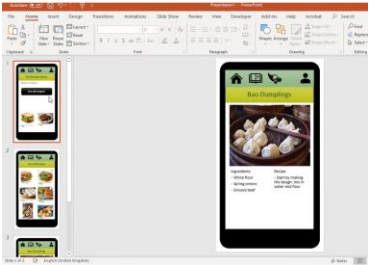


Remove slide from master view



Create home slide





Create links

Testing

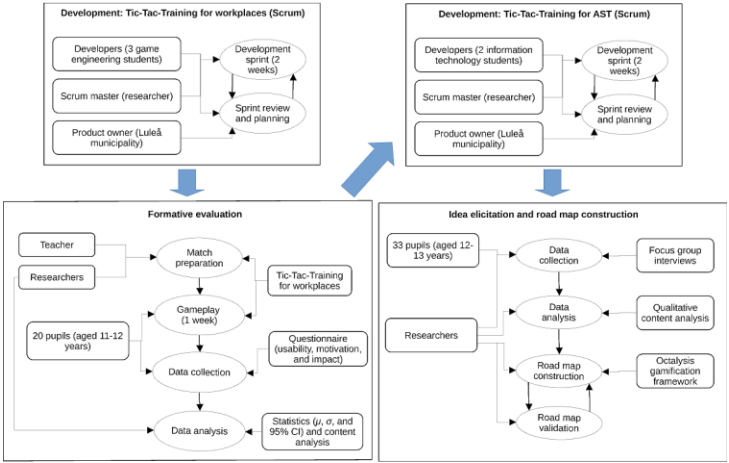
Slidedeck Example

Prototypes and Participatory Design

- Prototypes are an essential part of participatory design because people cannot always understand formal models or abstract representations but they can explore and evaluate ideas through engaging with prototyped systems.
- People can also be directly involved in prototype design.
- During the development of a prototype active school transport game, we ran a workshop with schoolchildren from a school in rural Sweden.

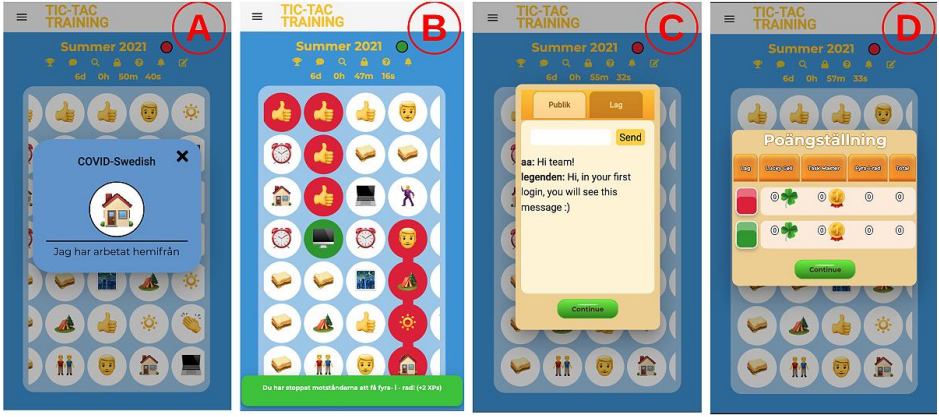
Example - Prototypes & Participatory Design

- Developed a game named Tic-Tac-Training for promoting active transport at workplaces.
- Applied Tic-Tac-Training as a reusable multiplayergame for promoting active school transport among schoolchildren in northern Sweden
- To meet the recommended daily physical activity levels of the World Health Organization



Laine, T. H., Duong, N., Lindvall, H., Oyelere, S. S., Rutberg, S., & Lindqvist, A. K. (2022). A reusable multiplayer game for promoting active school transport: development study. JMIR serious games, 10(1),

Tic-Tac-Training gameplay screens

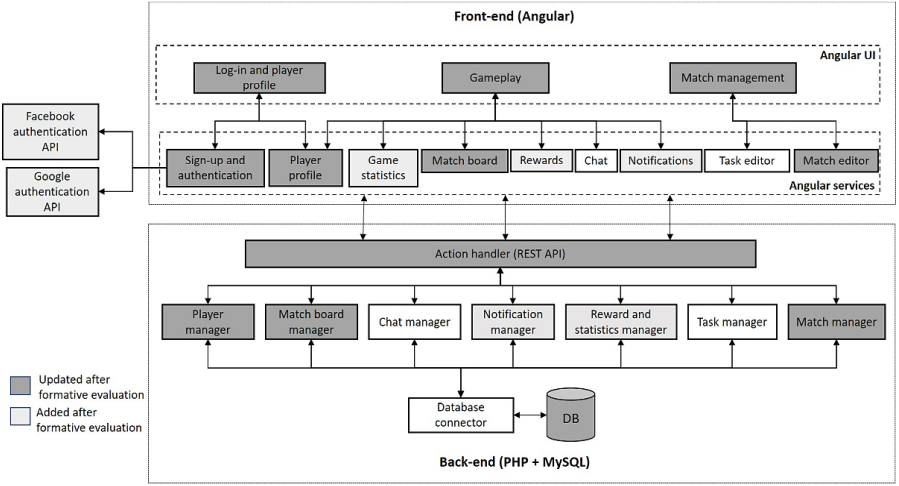


(A) Task details, (B) game board where the green team blocked the red team (C) in-game chat (D) match statistics

Tic-Tac-Training match and task creation screens

(A) Selecting how many teams will play; (B) assigning players to teams
(C) selecting tasks for the match;
(D) selecting the game board size;
(E) selecting distribution of the tasks
(F) choosing match name, period, ending condition, and chat;
(G) managing task categories;
(H) creating a new task;
(I) choosing an icon for a new task.

Tic-Tac-Training Architecture based on Angular, PHP Hypertext Preprocessor, & MySQL



Tic-Tac-Training New Features: Gameplay

(A) Lucky Cell reward,
(B) Task Master reward,
(C) level progression view,
(D) player statistics,
(E) notification settings,
(F) notification history,
(G) information about rewards in a match,
(H) information about how experience points can be earned in a match.

Prototyping Tools

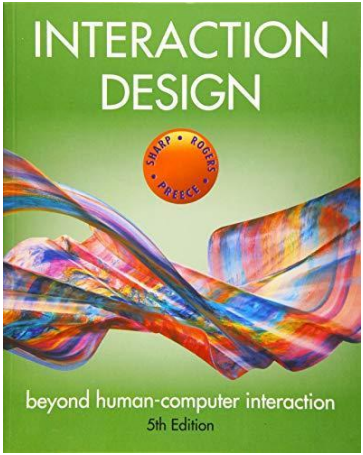
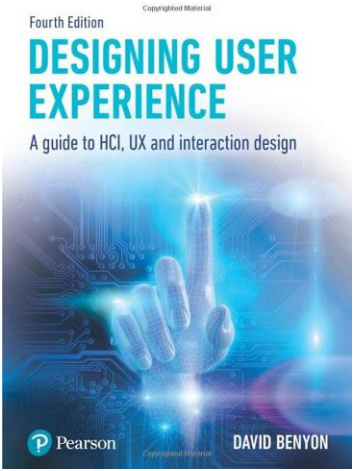
A good prototyping tool should:

- Allow easy, rapid modification of interface details or functionality.
- For designers who are not programmers, allow direct manipulation of prototype components.
- For incremental and evolutionary prototypes, facilitate reuse of code.
- Not constrain the designer to default styles for interface objects.
- Useful tools for requirements animation include paper, PowerPoint (e.g. for illustrating main screens) and drawing packages.
- Data manipulation languages such as SQL can be effective in animating the functionality of a system.

Prototyping Tools

- GUI design
  - Drawio
  - Balsamiq
  - FigJam
  - MockFlow
  - Moqups
  - Pen and Paper
  - PowerPoint
- Functionality
  - Wizard of Oz
  - Video
  - Simple code
  - Pen and Paper, Post-it notes, Storyboards

Balsamiq mockup

References

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